MATHEMATICS / INDEPENDENT STUDY

MM5020

Mathematics III - Abstract Algebra

Exercises and mock assessments

Original teaching material aligned with Stockholm University course descriptions.
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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

1 Unit 1 · Groups and cyclic structure

MM5020-U1.01 · E · 15 min
Learning objective: U1.1 Group axioms
Show that the even integers form a group under addition, and explain why they do not form a group under multiplication.

MM5020-U1.02 · E · 15 min
Learning objective: U1.1 Cancellation
Solve $axb = c$ in an arbitrary group and prove uniqueness.

MM5020-U1.03 · E · 15 min
Learning objective: U1.2 Element orders
Find the order and generated subgroup of $[9]$ in $\mathbb{Z}/15\mathbb{Z}$.

MM5020-U1.04 · E · 15 min
Learning objective: U1.2 Subgroup test
Prove that the intersection of two subgroups of a group is a subgroup.

MM5020-U1.05 · C · 30 min
Learning objective: U1.1 Structural identities
Prove that a group in which $x^2 = e$ for every x is abelian.

MM5020-U1.06 · C · 30 min
Learning objective: U1.2 Cyclic subgroups
List all subgroups of $\mathbb{Z}/18\mathbb{Z}$ and their orders.

MM5020-U1.07 · C · 30 min
Learning objective: U1.2 Generator arithmetic
Let a have order n. Prove $\langle a^r \rangle = \langle a^{\gcd(n,r)} \rangle$.

MM5020-U1.08 · C · 30 min
Learning objective: U1.2 Subgroup structure
Prove that the union of two subgroups H, K is a subgroup exactly when one contains the other.

MM5020-U1.09 · C · 30 min
Learning objective: U1.1 Noncommutative examples
Let r rotate a square by a quarter turn and s reflect it in a vertical axis. Show $srs = r^{-1}$ and use this to prove the symmetry group is nonabelian.

MM5020-U1.10 · A · 50 min
Learning objective: U1.2 Automorphisms
Classify all automorphisms of the additive group $\mathbb{Z}/n\mathbb{Z}$.

MM5020-U1.11 · A · 50 min
Learning objective: U1.2 Products of cyclic groups
Prove that $C_m \times C_n$ is cyclic exactly when $\gcd(m, n) = 1$.

MM5020-U1.12 · A · 50 min
Learning objective: U1.1 Axioms and finiteness
Let M be a finite nonempty set with an associative operation and a two-sided identity. Assume left and right cancellation. Prove that M is a group.

2 Unit 2 · Cosets and permutations

MM5020-U2.01 · E · 15 min
Learning objective: U2.1 Cosets
List the cosets of $H = \{[0], [4], [8]\}$ in $\mathbb{Z}/12\mathbb{Z}$.

MM5020-U2.02 · E · 15 min
Learning objective: U2.1 Lagrange
What element orders are possible in a group of order twenty-one according to Lagrange's theorem? Does this alone guarantee an element of order twenty-one?

MM5020-U2.03 · E · 15 min
Learning objective: U2.2 Permutation composition
Compute $(123)(23)$ and its sign, composing right to left.

MM5020-U2.04 · E · 15 min
Learning objective: U2.2 Cycle order
Find the order and sign of $(1234)(567)$ in S_7.

MM5020-U2.05 · C · 30 min
Learning objective: U2.1 Coset criteria
Prove $aH = bH$ if and only if $a^{-1}b \in H$.

MM5020-U2.06 · C · 30 min
Learning objective: U2.1 Prime order
Prove that a group of prime order has no proper nontrivial subgroup and is cyclic.

MM5020-U2.07 · C · 30 min
Learning objective: U2.2 Powers of cycles
Describe the disjoint cycles of $(123456)^2$ and $(123456)^3$, and compute their orders.

MM5020-U2.08 · C · 30 min
Learning objective: U2.2 Counting permutations
How many three-cycles are in S_5? How many products of two disjoint transpositions?

MM5020-U2.09 · C · 30 min
Learning objective: U2.2 Alternating groups
Show every three-cycle is even and every even permutation is a product of three-cycles.

MM5020-U2.10 · A · 50 min
Learning objective: U2.2 Minimal permutation degree
Find the least n for which S_n has an element of order fifteen.

MM5020-U2.11 · A · 50 min
Learning objective: U2.1 Indices in a tower
For finite G and $K \le H \le G$, prove $[G:K] = [G:H][H:K]$ by constructing coset representatives.

MM5020-U2.12 · A · 50 min
Learning objective: U2.2 Conjugation and cycles
Prove $\tau(a_1 \dots a_r)\tau^{-1} = (\tau(a_1) \dots \tau(a_r))$. Deduce that conjugate permutations have the same cycle lengths, and conversely.

3 Unit 3 · Homomorphisms and quotients

MM5020-U3.01 · E · 15 min
Learning objective: U3.1 Kernels
Find the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$, $f(n) = [3n]$.

MM5020-U3.02 · E · 15 min
Learning objective: U3.1 Normality
Show the kernel of the determinant map $\mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}^\times$ is normal.

MM5020-U3.03 · E · 15 min
Learning objective: U3.2 Quotient size
Let $|G| = 60$ and $N \lhd G$ have order twelve. Determine $|G/N|$ and its group structure.

MM5020-U3.04 · E · 15 min
Learning objective: U3.1 Homomorphisms
Show a homomorphism $f : G \to H$ satisfies $f(g^n) = f(g)^n$ for every integer n.

MM5020-U3.05 · C · 30 min
Learning objective: U3.2 First isomorphism theorem
Identify $\mathbb{R}^\times/\{1, -1\}$ using a natural homomorphism.

MM5020-U3.06 · C · 30 min
Learning objective: U3.1 Normality
Prove the center $Z(G)$ is a normal subgroup.

MM5020-U3.07 · C · 30 min
Learning objective: U3.2 Quotient maps
For positive integers m, n, when is $[a]_m \mapsto [a]_n$ well defined? When it is, find its kernel.

MM5020-U3.08 · C · 30 min
Learning objective: U3.2 Correspondence
List all subgroups of $\mathbb{Z}/12\mathbb{Z}$ by using subgroups of $\mathbb{Z}$ containing $12\mathbb{Z}$.

MM5020-U3.09 · C · 30 min
Learning objective: U3.1 Cyclic quotient by center
Prove that if $G/Z(G)$ is cyclic, then G is abelian.

MM5020-U3.10 · A · 50 min
Learning objective: U3.2 Universal property
Prove that a homomorphism $f : G \to H$ factors through G/N exactly when $N \subseteq \ker f$, and prove uniqueness of the factor.

MM5020-U3.11 · A · 50 min
Learning objective: U3.2 Abelianization
Let $[G, G]$ be generated by all commutators $aba^{-1}b^{-1}$. Show it is normal, its quotient is abelian, and every map from G to an abelian group factors uniquely through this quotient.

MM5020-U3.12 · A · 50 min
Learning objective: U3.1 Failure of converse to Lagrange
Prove that A_4 has no subgroup of order six. You may use that three-cycles generate A_4.

4 Unit 4 · Actions and simple groups

MM5020-U4.01 · E · 15 min
Learning objective: U4.1 Actions
Verify that S_n acts on $\{1, \dots, n\}$ by evaluation.

MM5020-U4.02 · E · 15 min
Learning objective: U4.1 Stabilizers
Find the stabilizer of one in the natural S_4 action and its order.

MM5020-U4.03 · E · 15 min
Learning objective: U4.2 Conjugacy
Compute $(123)(12)(123)^{-1}$.

MM5020-U4.04 · E · 15 min
Learning objective: U4.2 Simplicity
Which nontrivial finite cyclic groups are simple?

MM5020-U4.05 · C · 30 min
Learning objective: U4.1 Kernel of an action
Show the kernel of the homomorphism associated with an action is the intersection of all stabilizers.

MM5020-U4.06 · C · 30 min
Learning objective: U4.1 Conjugate stabilizers
Prove $G_{h \cdot x} = hG_x h^{-1}$.

MM5020-U4.07 · C · 30 min
Learning objective: U4.2 Class equation
Use conjugacy classes to show the center of S_3 is trivial.

MM5020-U4.08 · C · 30 min
Learning objective: U4.2 Prime-power groups
Prove a group of order p^2 is abelian.

MM5020-U4.09 · C · 30 min
Learning objective: U4.1 Orbit counting
Count two-colorings of four labeled square vertices up to rotation, with reflections excluded.

MM5020-U4.10 · A · 50 min
Learning objective: U4.2 Simple prime-power groups
Classify finite simple groups whose order is a power of a prime.

MM5020-U4.11 · A · 50 min
Learning objective: U4.1 Coset actions
Let H have finite index n in G. Show there is a normal subgroup $N \subseteq H$ with $[G : N] \leq n!$.

MM5020-U4.12 · A · 50 min
Learning objective: U4.2 Faithful actions of simple groups
Prove every nontrivial action of a simple group is faithful. Deduce that A_5 cannot act nontrivially on a set of four elements.

5 Unit 5 · Rings and their quotients

MM5020-U5.01 · E · 15 min
Learning objective: U5.1 Units
Find every unit in $\mathbb{Z}/10\mathbb{Z}$ and its inverse.

MM5020-U5.02 · E · 15 min
Learning objective: U5.1 Zero divisors
Give nonzero matrices $A, B \in M_2(\mathbb{Q})$ with $AB = 0$.

MM5020-U5.03 · E · 15 min
Learning objective: U5.2 Ideals
Show $6\mathbb{Z}$ is an ideal of $\mathbb{Z}$, and describe the identity of its quotient.

MM5020-U5.04 · E · 15 min
Learning objective: U5.1 Quaternions
Find the inverse of $q = 1 + 2i - j$.

MM5020-U5.05 · C · 30 min
Learning objective: U5.2 Quotient criteria
Determine whether (5) and (6) are prime or maximal in $\mathbb{Z}$.

MM5020-U5.06 · C · 30 min
Learning objective: U5.2 Evaluation
Show $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$.

MM5020-U5.07 · C · 30 min
Learning objective: U5.1 Characteristic
Prove a nonzero domain has characteristic zero or a prime.

MM5020-U5.08 · C · 30 min
Learning objective: U5.2 Ideals and units
Prove an ideal contains a unit if and only if it is the whole ring. Deduce that a field has exactly two ideals.

MM5020-U5.09 · C · 30 min
Learning objective: U5.2 Noncommutative ideals
Show the matrices with zero second column form a left ideal of $M_2(K)$ but not a two-sided ideal.

MM5020-U5.10 · A · 50 min
Learning objective: U5.2 Chinese remainder theorem
Prove $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ for coprime positive m, n.

MM5020-U5.11 · A · 50 min
Learning objective: U5.2 Matrix ring ideals
Prove every two-sided ideal of $M_{n(K)}$, for a field K, is zero or the whole ring.

MM5020-U5.12 · A · 50 min
Learning objective: U5.1 Division rings
Prove quaternion conjugation reverses products and deduce multiplicativity of $N(q) = q\overline{q}$.

6 Unit 6 · Polynomials and factorization

MM5020-U6.01 · E · 15 min
Learning objective: U6.1 Division
Divide $x^3 + 2x + 1$ by $x^2 + 1$ over $\mathbb{Q}$.

MM5020-U6.02 · E · 15 min
Learning objective: U6.1 Bézout
Find integers u, v with $30u + 21v = 3$.

MM5020-U6.03 · E · 15 min
Learning objective: U6.2 Irreducibility
Prove $x^3 - 5$ is irreducible over $\mathbb{Q}$.

MM5020-U6.04 · E · 15 min
Learning objective: U6.1 Roots
Factor $x^2 + 1$ over $\mathbb{F}_5$.

MM5020-U6.05 · C · 30 min
Learning objective: U6.1 Modular inverses
Find the inverse of $x + 1$ in $\mathbb{Q}[x]/(x^2 + 1)$.

MM5020-U6.06 · C · 30 min
Learning objective: U6.2 Reduction test
Prove $x^3 + x + 1$ is irreducible over $\mathbb{Q}$ by reducing modulo two.

MM5020-U6.07 · C · 30 min
Learning objective: U6.2 Failure outside PIDs
Show 2 is irreducible but not prime in $\mathbb{Z}\left[\sqrt{-5}\right]$.

MM5020-U6.08 · C · 30 min
Learning objective: U6.1 Principal ideals
Show $K[x]/(f)$ is a field if and only if f is irreducible, assuming f is nonconstant.

MM5020-U6.09 · C · 30 min
Learning objective: U6.2 Primitive products
Show the product of two primitive integer polynomials is primitive.

MM5020-U6.10 · A · 50 min
Learning objective: U6.2 Shifted Eisenstein
For prime p, prove $1 + x + \ldots + x^{p-1}$ is irreducible over $\mathbb{Q}$.

MM5020-U6.11 · A · 50 min
Learning objective: U6.1 Polynomial Chinese remainder
If $f, g \in K[x]$ are coprime, prove $K[x]/(fg) \cong K[x]/(f) \times K[x]/(g)$.

MM5020-U6.12 · A · 50 min
Learning objective: U6.2 Nonprincipal ideals
Prove $\mathbb{Z}[x]$ is a UFD but not a PID.

7 Unit 7 · Fields as vector spaces

MM5020-U7.01 · E · 15 min
Learning objective: U7.1 Fractions
Prove $\frac{a}{b} = 0$ in the fraction field of a domain exactly when $a = 0$.

MM5020-U7.02 · E · 15 min
Learning objective: U7.2 Minimal polynomials
Find the minimal polynomial of $\sqrt{7}$ over $\mathbb{Q}$ and a basis of its generated field.

MM5020-U7.03 · E · 15 min
Learning objective: U7.1 Tower law
If $[L : K] = 3$ and $[M : L] = 4$, find $[M : K]$. Can an intermediate field between K and M have degree five over K?

MM5020-U7.04 · E · 15 min
Learning objective: U7.2 Power-basis arithmetic
With $\alpha^3 = 2$, simplify $\alpha^5 + \alpha^4$ in the power basis.

MM5020-U7.05 · C · 30 min
Learning objective: U7.2 Inverses
Find $\left(2 + \sqrt{3}\right)^{-1}$ in $\mathbb{Q}\left(\sqrt{3}\right)$ and explain why every nonzero $a + b\sqrt{3}$ has an inverse in this field.

MM5020-U7.06 · C · 30 min
Learning objective: U7.2 Algebraicity
Prove every element of a finite extension is algebraic.

MM5020-U7.07 · C · 30 min
Learning objective: U7.2 Change of generator
Find the minimal polynomial of $1+\sqrt{2}$ over $\mathbb{Q}$.

MM5020-U7.08 · C · 30 min
Learning objective: U7.1 Fraction universal property
Explain why the reduction map $\mathbb{Z} \to \mathbb{F}_5$ cannot extend to a unital homomorphism $\mathbb{Q} \to \mathbb{F}_5$.

MM5020-U7.09 · C · 30 min
Learning objective: U7.2 Root adjunction
Construct a field containing a root of x^3+x+1 over $\mathbb{F}_2$ and list a basis over $\mathbb{F}_2$.

MM5020-U7.10 · A · 50 min
Learning objective: U7.1 Compositum degree
Prove $\left[\mathbb{Q}\left(\sqrt{2},\sqrt{5}\right):\mathbb{Q}\right]=4$ and find a basis.

MM5020-U7.11 · A · 50 min
Learning objective: U7.2 Transitivity of algebraicity
Let $\frac{L}{K}$ be algebraic and let α be algebraic over L. Prove α is algebraic over K, even when $\frac{L}{K}$ is infinite.

MM5020-U7.12 · A · 50 min
Learning objective: U7.2 A primitive generator
Show $\mathbb{Q}\left(\sqrt{2}+\sqrt{3}\right)=\mathbb{Q}\left(\sqrt{2},\sqrt{3}\right)$ and find the minimal polynomial of the sum.

8 Unit 8 · Finite fields and applications

MM5020-U8.01 · E · 15 min
Learning objective: U8.1 Field orders
Which of the orders $8, 9, 10, 16, 18, 25$ can occur for a finite field?

MM5020-U8.02 · E · 15 min
Learning objective: U8.1 Arithmetic in four elements
In $\mathbb{F}_4 = \mathbb{F}_2[\alpha]$ with $\alpha^2 + \alpha + 1 = 0$, compute $\alpha(\alpha + 1)$ and the inverse of $\alpha + 1$.

MM5020-U8.03 · E · 15 min
Learning objective: U8.2 Generators
Find the order of two in $\mathbb{F}_7^{\times}$.

MM5020-U8.04 · E · 15 min
Learning objective: U8.1 Characteristic and dimension
What are the characteristic and dimension over the prime field of a field of order eighty-one?

MM5020-U8.05 · C · 30 min
Learning objective: U8.2 Frobenius
Prove the map $a \mapsto a^p$ is an automorphism of a finite field of characteristic p.

MM5020-U8.06 · C · 30 min
Learning objective: U8.2 Interpolation
Find the degree-less-than-three polynomial over $\mathbb{F}_5$ with $f(0) = 1$, $f(1) = 0$, $f(2) = 2$.

MM5020-U8.07 · C · 30 min
Learning objective: U8.1 Polynomial roots
Show every element of $\mathbb{F}_q$ is a root of $x^q - x$, and this polynomial has no repeated root.

MM5020-U8.08 · C · 30 min
Learning objective: U8.2 Evaluation codes
Evaluate all degree-less-than-two polynomials at the five elements of $\mathbb{F}_5$. How many codewords result, and what is their minimum distance?

MM5020-U8.09 · C · 30 min
Learning objective: U8.2 Primitive elements
Find all generators of $\mathbb{F}_7^\times$, given that three is one generator.

MM5020-U8.10 · A · 50 min
Learning objective: U8.1 Subfields
Show a field of sixteen elements has a unique subfield of four elements.

MM5020-U8.11 · A · 50 min
Learning objective: U8.2 Polynomial functions
Prove every function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a unique polynomial of degree less than q. Is the representing polynomial unique without the degree bound?

MM5020-U8.12 · A · 50 min
Learning objective: U8.1 Irreducible quadratics
Count the monic irreducible quadratics over $\mathbb{F}_q$.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M1.01 · E · 15 min
Learning objective: U1–U2 Cyclic groups
List the orders of every element of $\mathbb{Z}/8\mathbb{Z}$.

MM5020-M1.02 · E · 15 min
Learning objective: U5 Ring quotients
Determine whether $\mathbb{Z}/15\mathbb{Z}$ is a field and list its units.

MM5020-M1.03 · C · 30 min
Learning objective: U3–U4 Group structure
Let $|G| = 49$. Prove G is abelian, and show that if some element has order forty-nine then G is cyclic.

MM5020-M1.04 · C · 30 min
Learning objective: U6–U7 Algebraic extensions
Let $\alpha^3 = 3$ over $\mathbb{Q}$. Prove the extension degree is three and find $(\alpha - 1)^{-1}$.

MM5020-M1.05 · A · 50 min
Learning objective: U3 Universal constructions
For normal subgroups N, M of G, show $g \mapsto (gN, gM)$ induces an injective homomorphism $G/(N \cap M) \to G/N \times G/M$. When is it surjective?

MM5020-M1.06 · A · 50 min
Learning objective: U8 Field structure
Prove that the fixed field of Frobenius $x \mapsto x^p$ in $\mathbb{F}_{p^n}$ is exactly $\mathbb{F}_p$.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M2.01 · E · 15 min
Learning objective: U2 Permutations
Find the order and sign of $(123)(45)(67)$.

MM5020-M2.02 · E · 15 min
Learning objective: U7 Minimal polynomials
Find the minimal polynomial of $2+\sqrt{5}$ over $\mathbb{Q}$.

MM5020-M2.03 · C · 30 min
Learning objective: U3 Quotient isomorphisms
Prove $S_4/A_4 \cong C_2$.

MM5020-M2.04 · C · 30 min
Learning objective: U5–U6 Field construction
Prove $\mathbb{F}_3[x]/(x^2+1)$ is a field and find the inverse of $[x+1]$.

MM5020-M2.05 · A · 50 min
Learning objective: U4 Actions
A group G of order fifteen acts on a set of seven elements. Prove there is a fixed point.

MM5020-M2.06 · A · 50 min
Learning objective: U7–U8 Degrees and subfields
Prove that $\mathbb{F}_{p^3}$ has no subfield strictly between $\mathbb{F}_p$ and itself.